Number Of Columns : 70

Text Areas : PlainText

Question Number : 33 Question Id : 640653446373 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2.25

Question Label : Short Answer Question

Mention some suggestions to improve the program.

NOTE: Your answer should not exceed 300 words.

Response Type : Alphanumeric

Evaluation Required For SA : No

Max Word Count : 300

Show Word Count : Yes

Min Word Count : 0

Highlight min word : Yes

Single Line Response : No

Number of Rows : 10

Number Of Columns : 70

Text Areas : PlainText

Speech Tech

Section Id :	64065329021
Section Number :	2
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	3
Number of Questions to be attempted :	3
Section Marks :	25

Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065363599
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 34 Question Id : 640653446374 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL: SPEECH TECHNOLOGY"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531487009. ✔ Yes

6406531487010. ✖ No

Sub-Section Number :	2
Sub-Section Id :	64065363600
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 35 Question Id : 640653446375 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 12

Question Label : Multiple Choice Question

1. (1 point) Consider a sinlge perceptron $f(z)$ $[z = W^T x,\ f(z) = 1 \text{ if } z \geq \theta]$ having 4 inputs $x = [2, 1, 1, 0]^T$ and 4 weights $W = [0.05, 0.7, 0.7, 0.3]^T$ and thresold θ. Given that the threshold $\theta = 1.2$ will the perceptron get activated?

2. (2 points) State true or False: Non-Linearly separable problems can be solved by single perceptron. Give reasons for your answer.

3. (5 points) Let us assume we want to implement an XOR function using MLP. Below is a tabular representation of XOR function:

X_1	X_2	$X_1 \oplus X_2$
0	0	0
0	1	1
1	0	1
1	1	0

The activation function of a single neuron is denoted as:

$$f(x) = \begin{cases} 0, & \text{for } x < 0 \\ 1, & \text{for } x \geq 0 \end{cases}$$

Model this function using an MLP.

4. (3 points) The sigmoid function is given as $f(z) = \frac{1}{1+\exp^{-z}}$. Compute the mathematical expression for the derivative of $f(z)$.

5. (1 point) Which CTC alignment corresponds to the text "speech"?

   A. —sspppee-ch

   B. -sppp—eech

   C. -s-pp–e-ee-ccch

   D. —spppee–ch

Options :

6406531487011. ✔ I have written answers on the answer sheets

6406531487012. ✖ Not applicable

Sub-Section Number :	3
Sub-Section Id :	64065363601
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 36 Question Id : 640653446376 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 13

Question Label : Multiple Choice Question

6. (1 point) Name few applications where CTC is used.

7. (2 points) Suppose that in the training corpus, the word sequences "you" occurred 500 times, "How are you" occurred 300 times and "How are" occurred 800 times, what is the probability that the word "you" follows the word sequence "How are"?

8. (2 points) Explain in brief the difference between Markov Chains and Hidden Markov models.

9. (5 points) Consider a weather prediction based HMM problem with three hidden states: *Sunny, Cloudy and Rainy.* There are four different observable states that depend on the hidden states, namely: *Dry, Dryish, Damp, and Soggy.* The transition and observation probabilities are shown in fig. Given initial Cloudy weather condition on Day0. It was observed that the weather was *Dry* followed by *Damp* for the next two days. Compute the best possible weather forecast sequence which best explains these two observations (*hint: Use Viterbi Algorithm*).

Grass

Weather	Dry	Dryish	Damp	Soggy
Sunny	0.60	0.20	0.15	0.05
Cloudy	0.25	0.25	0.25	0.25
Rainy	0.05	0.10	0.35	0.50

Today

Yesterday	Sunny	Cloudy	Rainy
Sunny	0.5	0.375	0.125
Cloudy	0.25	0.125	0.625
Rainy	0.25	0.075	0.675

10. (3 points) Explain in detail different types of HMM models detailing differences between them.

Options :

6406531487013. ✔ I have written answers on the answer sheets

6406531487014. ✖ Not applicable

Section Id :	64065329022
Section Number :	3
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	7
Number of Questions to be attempted :	7
Section Marks :	20
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	64065363602
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 37 Question Id : 640653446377 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL: INDUSTRY 4.0"

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CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531487015. ✔ Yes

6406531487016. ✖ No

Sub-Section Number :	2